$$= \frac{1}{2a}[\log|x-a| - \log|x+a|] + c$$

$$= \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + c$$

Therefore $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + c$

(iii) Let $I = \int \frac{dx}{a^2+x^2}$.

Putting $x = a\tan\theta \Rightarrow \theta = \tan^{-1}\frac{x}{a}$

$dx = a\sec^2\theta$

$$I = \int \frac{a\sec^2\theta}{a^2+a^2\tan^2\theta}d\theta = \int \frac{a\sec^2\theta}{a^2(1+\tan^2\theta)}d\theta = \int \frac{a\sec^2\theta}{a^2\sec^2\theta}d\theta = \frac{1}{a}\int d\theta$$

$$= \frac{1}{a}\theta + c = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$$

Therefore, $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$

(iv) Let $I = \int \frac{dx}{\sqrt{a^2-x^2}}$.

Putting $x = a\sin\theta \Rightarrow \theta = \sin^{-1}\left(\frac{x}{a}\right)$

$dx = a\cos\theta$

$$I = \int \frac{a\cos\theta}{\sqrt{a^2-a^2\sin^2\theta}}d\theta = \int \frac{a\cos\theta}{\sqrt{a^2(1-\sin^2\theta)}}d\theta = \int \frac{a\cos\theta}{a\cos\theta}d\theta = \int d\theta$$

$$= \theta + c = \sin^{-1}\left(\frac{x}{a}\right) + c$$

Therefore, $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + c$

(v) Let $I = \int \frac{dx}{\sqrt{x^2-a^2}}$

Putting $x = a\sec\theta \Rightarrow \theta = \sec^{-1}\left(\frac{x}{a}\right)$

$dx = a\sec\theta\tan\theta d\theta$

$$I = \int \frac{a\sec\theta\tan\theta}{\sqrt{a^2\sec^2\theta - a^2}}d\theta = \int \frac{a\sec\theta\tan\theta}{\sqrt{a^2(\sec^2\theta-1)}}d\theta = \int \frac{a\sec\theta\tan\theta}{a\tan\theta}d\theta = \int \sec\theta d\theta$$

$$= \log|\sec\theta + \tan\theta| + c$$

11_11th-Maths-Vol2-Unit11.indd 215 16-12-2022 11:07:51